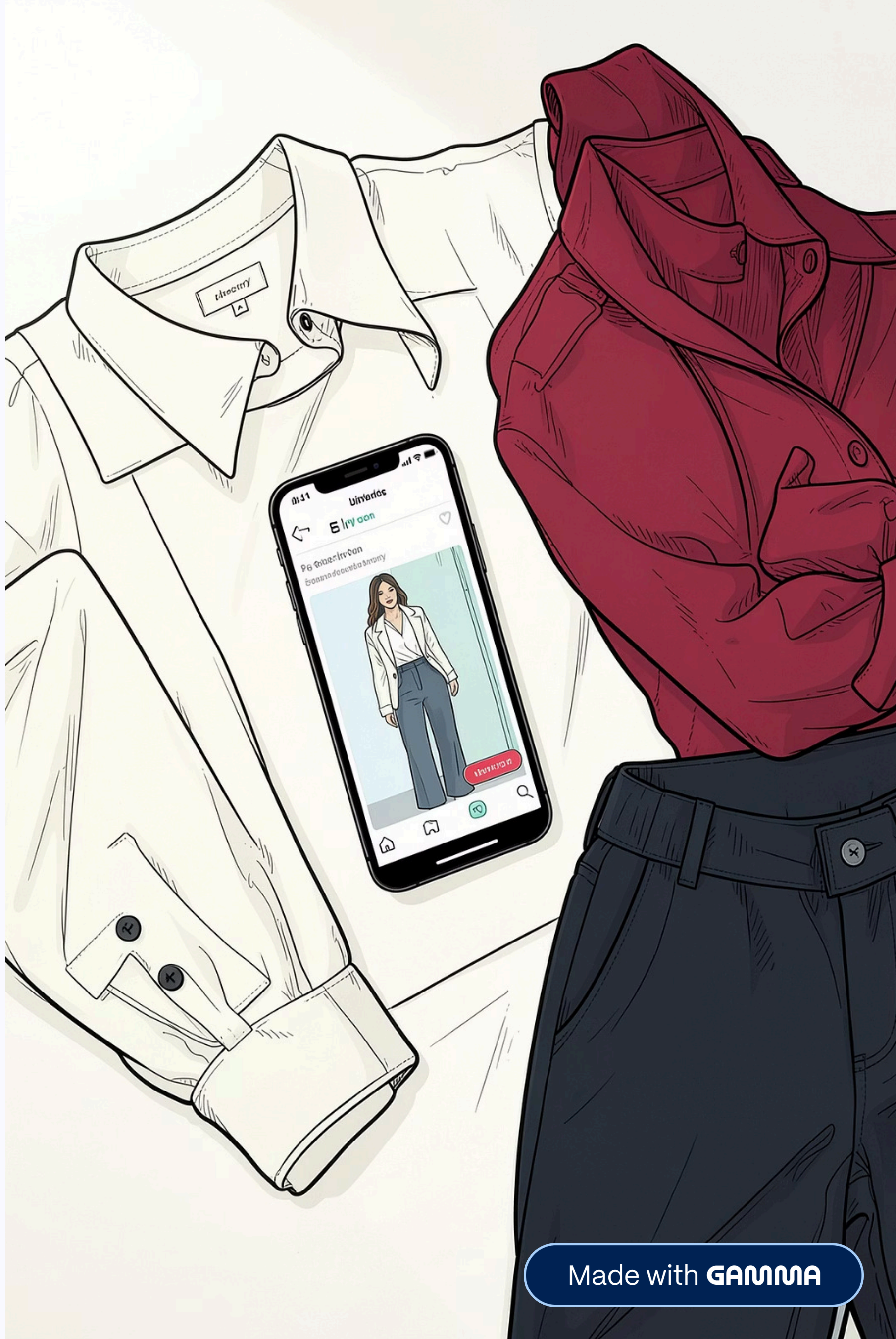
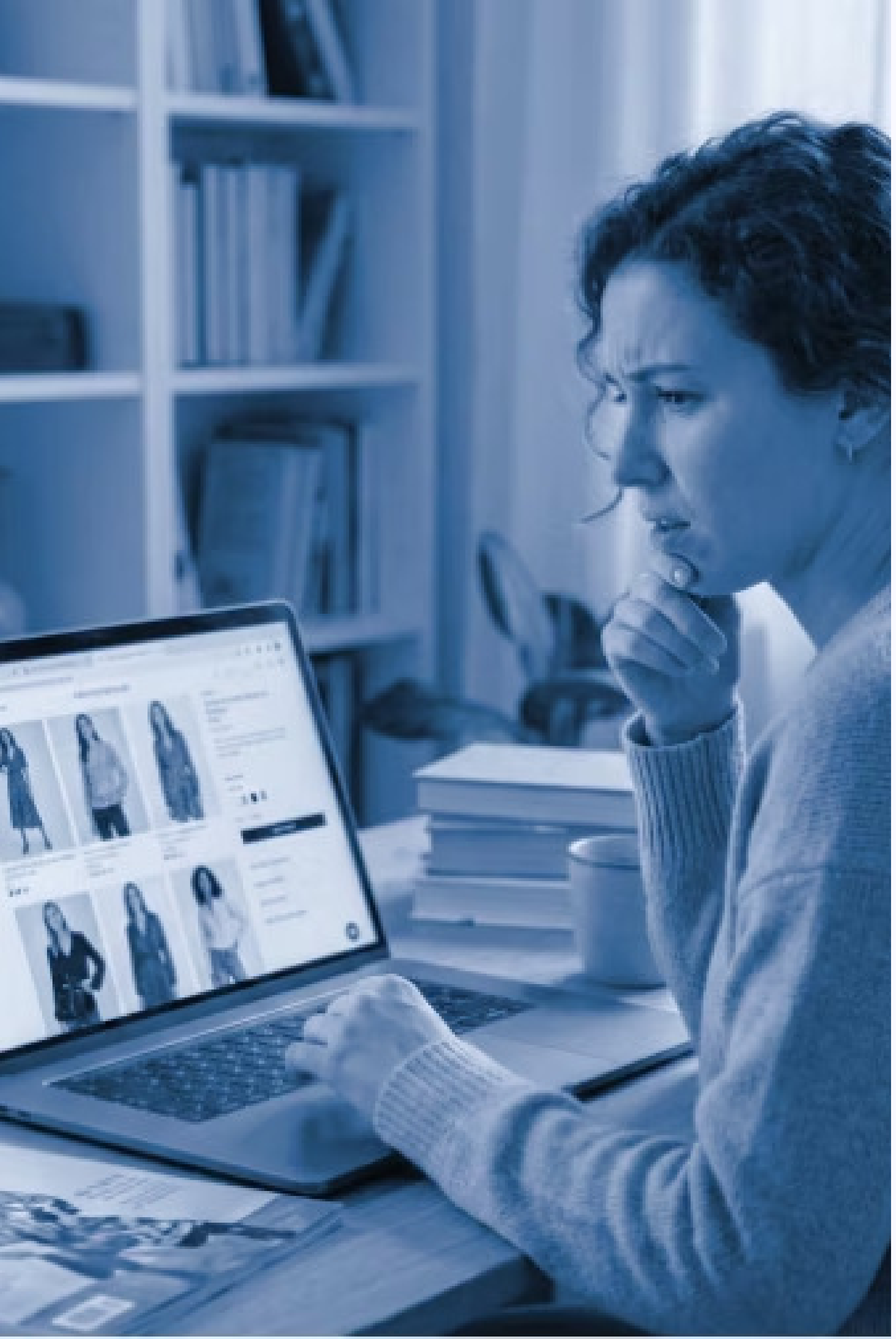
AI-Powered Virtual Outfit Try-On

Human - AI Interaction · FCUL, University of Lisbon

Rafael Ventura, Rodrigo Lopes, Gonçalo Vieira e Pedro Neves



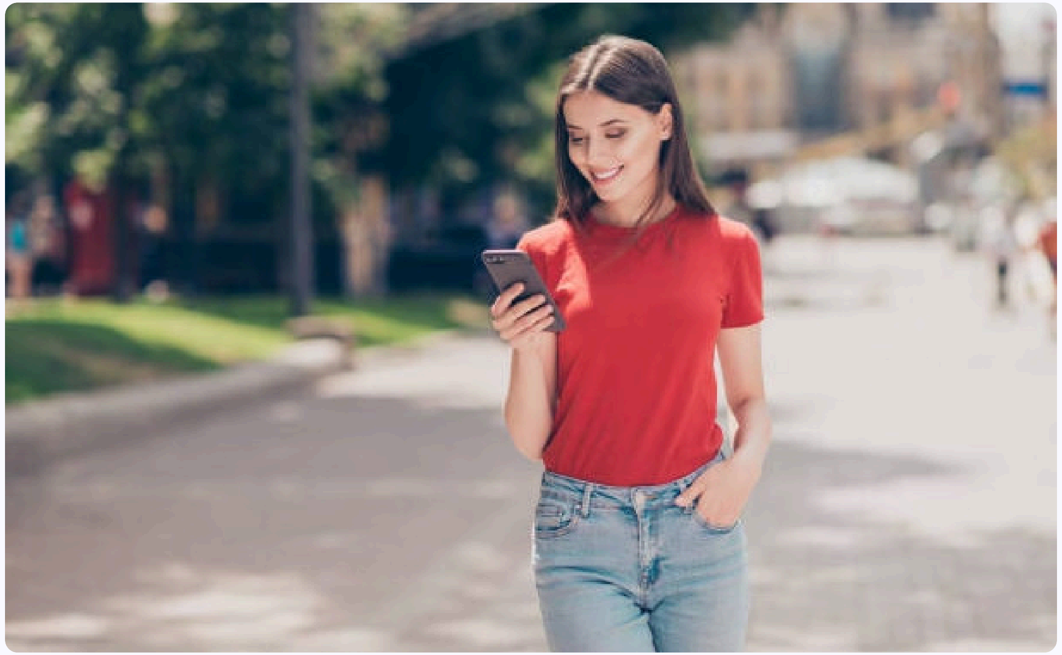


Problem Description

The adoption of online fashion retail websites remains limited due to the difficulty users face in accurately perceiving how clothes will fit on their bodies.

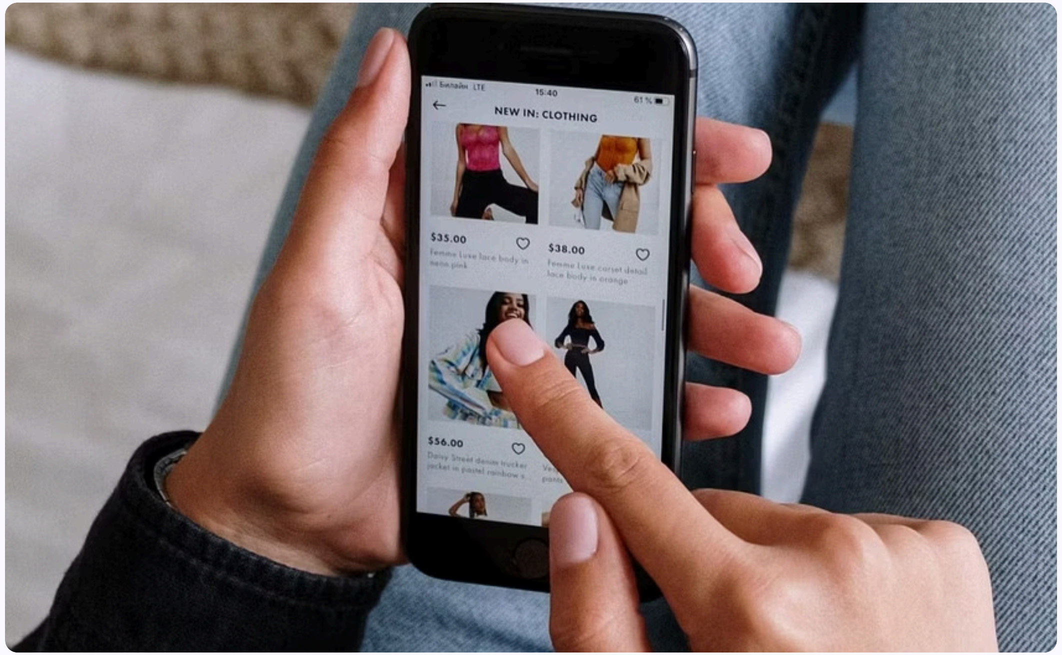
According to a 2019 study [1], website quality significantly impacts user trust and perceived utility, indicating that more advanced and informative interfaces can reduce the risks associated with online shopping. Additionally, user experience is the most influential factor in determining whether a user will continue using the system.

Our Target Users



Young People

Digitally native, style-conscious, and accustomed to personalized experiences. They expect tools that reflect their identity, instead of generic recommendations.



Online Shoppers

Anyone who buys clothes online and has experienced the frustration of returns, sizing mismatches, or disappointing deliveries.

The User Journey / Storyboard

A three-step intake flows into AI-generated recommendations and a personalized try-on image. Returning users are remembered via browser storage, so no account is required.

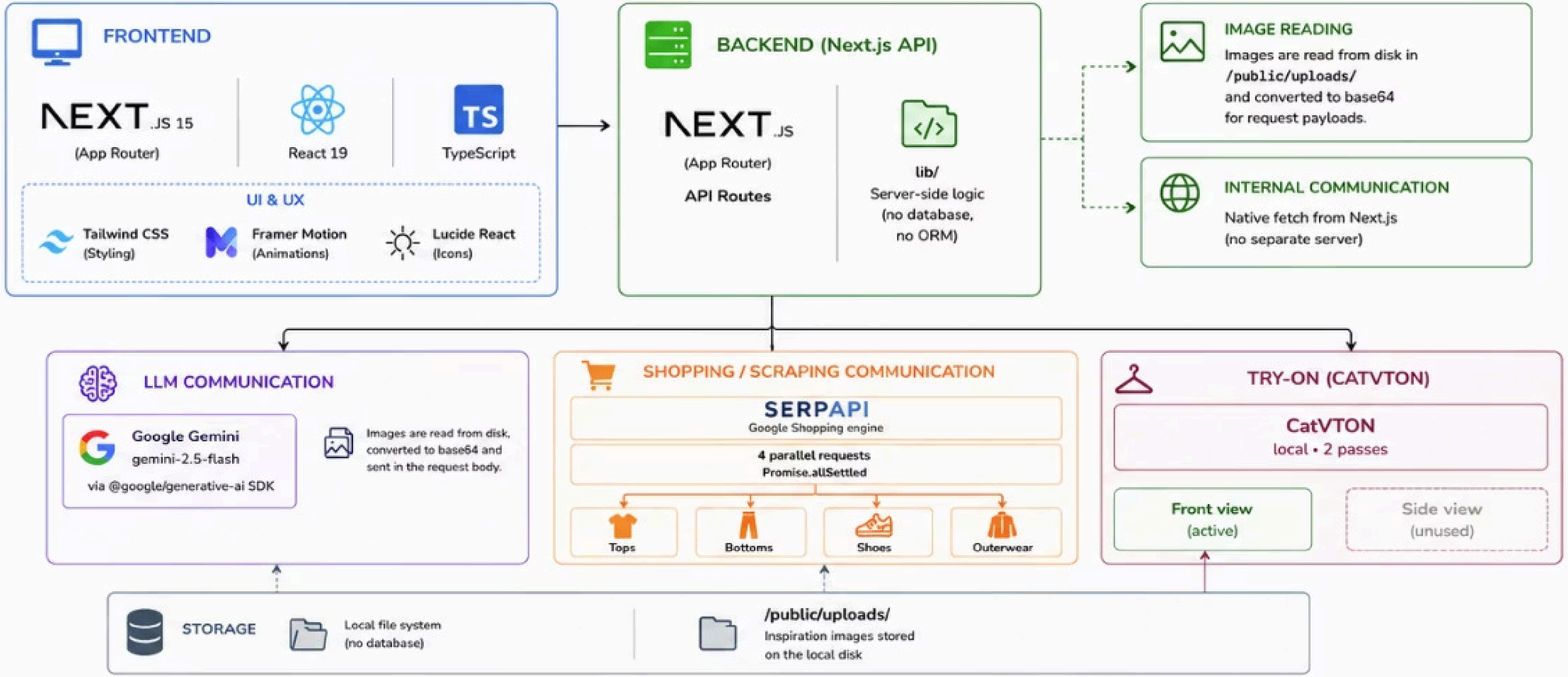
Data Flow & Storage

What the App Stores

- Physical attributes: height, weight, body type, clothing sizes
- Portrait image (uploaded or preset)
- Style selection and free-text notes
- Up to two reference photos
- Full prompts sent to Gemini

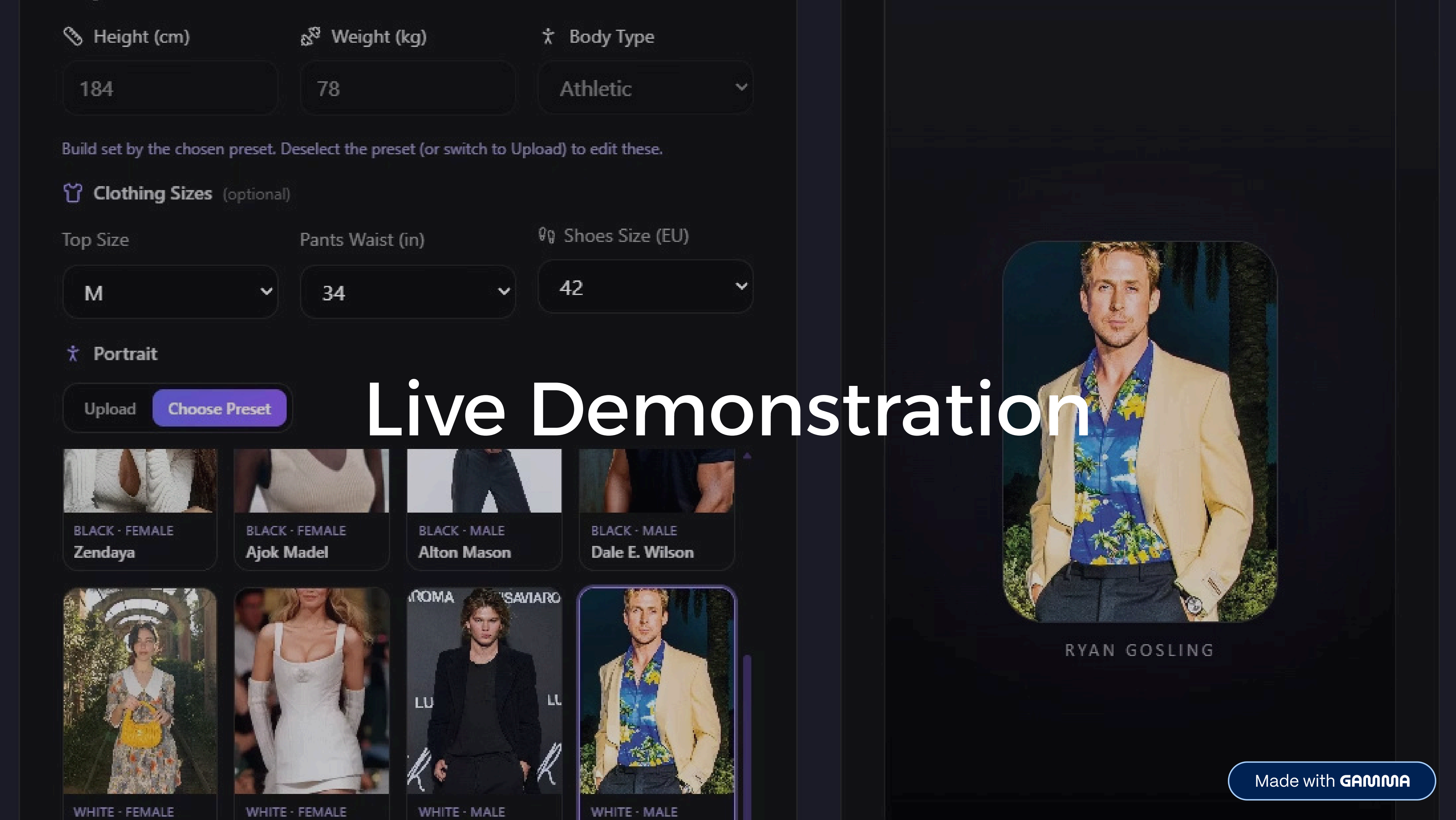
> As this is a prototype, all data is stored in browser cache. In production, a database would be needed to ensure persistence, scaling and security.

System Architecture



A clean separation of concerns: the frontend captures intent, the backend coordinates three specialized AI services, and all session data lives in the browser, keeping the prototype lightweight (apart from CatVTON) and account-free.

Live Demonstration



User Study Design

Methodology

To develop a survey aimed at understanding user interactions and experiences with the AI-enabled system, we followed guidelines established in the state-of-the-art literature [2], including some from the medical field [3] [4]. 18 respondents completed the study after interacting with the prototype.

Main Focus Areas

- Perceived accuracy and reliability of AI output
- Match between results and user expectations
- Transparency and ease of accessing explanations

Survey Results: What Users Told Us

64%

Clarity of capabilities

Users understood what the system could do

38%

Awareness of AI errors

Users understood how often the AI makes mistakes

74%

Relevant information

Users felt the information was useful and relevant

89%

Fair and unbiased

Users viewed the system as fair and unbiased

78%

Privacy respected

Users felt their privacy was respected

58%

Predictable behavior

Users could easily understand why the system behaved a certain way

Respondents answered each item on a 1-5 rating scale or a yes/no/maybe choice. Ratings were mapped linearly from 1=0% to 5=100%, and choices were mapped no/maybe/yes to 0/50/100%. Each percentage is the average across all respondents for that item. The survey had a total of 18 participants.

What UseSurvey Results: What Users Told Us

83%

Ease of feedback to AI

Users found feedback easy to provide

58%

Met expectations

Results aligned with user expectations

64%

Wanted results

Users got the results they wanted

61%

Would return

Users would use the system again

56%

Would recommend

Users would recommend the system to a friend or family member

Respondents answered each item on a 1-5 rating scale or a yes/no/maybe choice. Ratings were mapped linearly from 1=0% to 5=100%, and choices were mapped no/maybe/yes to 0/50/100%. Each percentage is the average across all respondents for that item. The survey had a total of 18 participants.

Discussion & Future Work

✅ Solid Foundation

After analyzing user feedback, we can conclude that the prototype of this AI-enabled system has a solid foundation, demonstrating that the interactions between the user and the AI models were well optimized.

🗄️ Real Database

Replace browser storage with a proper backend to enable persistence, cross-device use, and richer user history.

📐 True 3D Modeling

Move beyond 2D portrait layering toward full image-to-3D body modeling for more accurate and dynamic try-on results.

🔍 Expand & Explain

Broaden the preset portrait gallery and improve system transparency so users better understand how recommendations are generated.

Thank You

Human - AI Interaction · FCUL, University of Lisbon

Rafael Ventura, Rodrigo Lopes, Gonçalo Vieira e Pedro Neves

References

[1] Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., ... Horvitz, E. (2019). Guidelines for Human-AI Interaction. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 1 - 13. Presented at the Glasgow, Scotland Uk. doi:10.1145/3290605.3300233

[2] Ji, M., Genchev, G. Z., Huang, H., Xu, T., Lu, H., & Yu, G. (2021). Evaluation framework for successful artificial intelligence - enabled clinical decision support systems: mixed methods study. Journal of medical Internet research, 23(6), e25929.

[3] Reddy, S., Rogers, W., Makinen, V. P., Coiera, E., Brown, P., Wenzel, M., ... & Kelly, B. (2021). Evaluation framework to guide implementation of AI systems into healthcare settings. BMJ health & care informatics, 28(1), e100444.